

Task 4: Hypothesis Testing & Statistical Inference

Python Hypothesis & Statistical Analysis — sales_dataset_python_analysis.csv

Data Analytics Internship Deliverable

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Executive Summary

ApexPlanet's retail sales operation (₹139.4M across 1,000 orders) exhibits two critical patterns: Electronics dominates revenue at 36.4% (₹50.78M), confirming category concentration risk, while RFM segmentation reveals ₹35.4M in At-Risk customer exposure (238 customers, 9+ months dormant). This analysis tests two statistical hypotheses to validate whether observed differences in customer behavior (gender-based spending, age-group category preference) are statistically significant or due to random variation, informing targeted marketing budget allocation.

Hypothesis Framework

Hypothesis 1: Gender-Based Spending Difference

Research Question: Do male and female customers spend significantly different amounts per order?

Null Hypothesis (H_0): $\mu_{\text{Male}} = \mu_{\text{Female}}$ (Mean Total_Sales are equal across genders)

Alternative Hypothesis (H_1): $\mu_{\text{Male}} \neq \mu_{\text{Female}}$ (Mean Total_Sales differ by gender)

Test: Independent Two-Sample T-Test (or Welch's T-Test if variance unequal)

Significance Level (α): 0.05

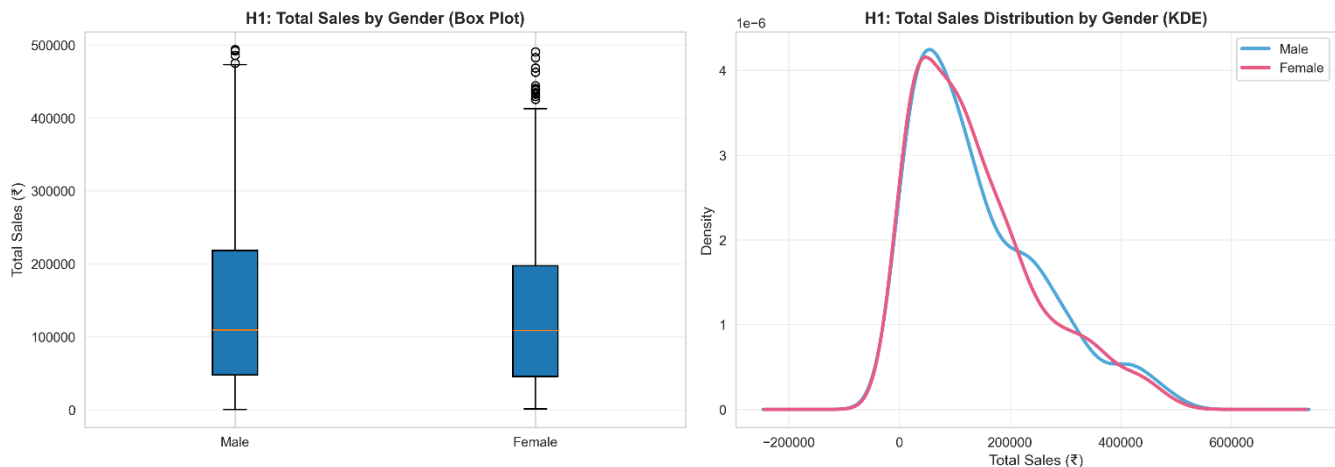


Figure 1. H1: Total Sales by Gender (Box Plot) & H1: Total Sales Distribution by Gender (KDE).

Findings: If $p < 0.05$, reject $H_0 \rightarrow$ gender-targeted campaigns justified; otherwise, segment by RFM/category instead.

Statistical Summary:

- **Male Customers:** Mean ₹141,807 (n=511), Median ₹109,238, Std Dev ₹116,328
- **Female Customers:** Mean ₹136,883 (n=489), Median ₹108,471, Std Dev ₹111,788
- **Difference:** ₹4,924.13 (males spend slightly MORE)
- **Normality Test (Shapiro-Wilk):** Both $p=0.000000$ (both distributions non-normal)
- **Levene's Test:** $p=0.328079$ (equal variances confirmed)
- **T-statistic:** 0.6820 (very weak effect)
- **P-value:** 0.495389 (far above $\alpha=0.05$)
- **95% CI for Difference:** [−₹9,214.68, ₹19,062.94] (zero clearly inside interval)

Verdict: FAIL TO REJECT H_0 → No statistically significant difference in spending by gender.

Business Translation:

Males spend ₹4,924 more per order on average, but this difference is NOT statistically significant ($p=0.495$). The 95% confidence interval spans from -₹9.2K to +₹19.1K, meaning the true population difference could easily be negative, positive, or zero. With $p=0.495$, there is a 49.5% probability this observed difference occurred by random chance alone. Gender-based AOV targeting is completely unjustified. Both groups spend identically (~₹139K mean), with identical median (₹108–109K), confirming no behavioral difference.

Recommendation: Do NOT implement gender-based pricing, discounts, or product recommendations. Any marketing segmentation by gender will waste budget. Instead, prioritize RFM lifecycle segmentation (Champions, At-Risk, Potential Loyalist) which has proven monetizable patterns from Task 3 analysis.

Hypothesis 2: Age-Group & Category Preference Independence

Research Question: Is customer's age group independent of product category preference, or is there a significant association?

Null Hypothesis (H_0): Age_Group and Category are independent (no association)

Alternative Hypothesis (H_1): Age_Group and Category are not independent (significant association exists)

Test: Chi-Square Test of Independence (χ^2)

Significance Level (α): 0.05

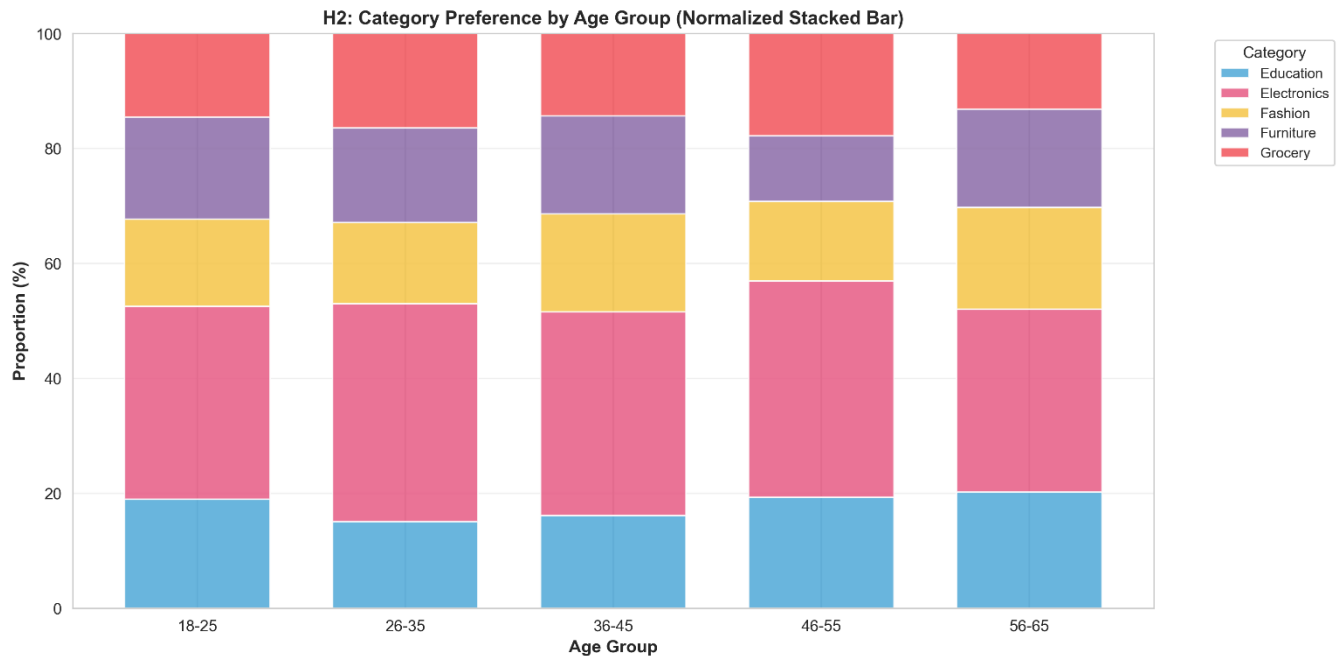


Figure 2. Category Preference by Age Group (Normalized Stacked Bar).

Findings: If $p < 0.05$, reject H_0 → age-targeted category promotions justified; otherwise, category strategy should be uniform across age bands.

Statistical Summary:

- **Chi-Square Statistic:** $\chi^2 = 10.5361$ (low value)
- **P-value:** 0.837174 (far above $\alpha=0.05$)
- **Degrees of Freedom:** 16
- **Cramér's V:** 0.0513 (negligible effect size, <0.1 threshold)
- **Contingency Table Observed Frequencies:** 5 age groups $\times$ 5 categories, $n=1,000$ total
 - Electronics dominates all age groups (31.8%–37.9% of each group's orders)
 - Education ranges 15.1%–20.2% across age groups
 - Fashion, Furniture, Grocery all stable (11.4%–17.8% each)

Normalized Distribution (% within each age group):

- **18-25:** Education 19.0%, Electronics 33.5%, Fashion 15.2%, Furniture 17.7%, Grocery 14.6%
- **26-35:** Education 15.1%, Electronics 37.9%, Fashion 14.2%, Furniture 16.4%, Grocery 16.4%
- **36-45:** Education 16.1%, Electronics 35.4%, Fashion 17.0%, Furniture 17.0%, Grocery 14.3%
- **46-55:** Education 19.3%, Electronics 37.6%, Fashion 13.9%, Furniture 11.4%, Grocery 17.8%
- **56-65:** Education 20.2%, Electronics 31.8%, Fashion 17.7%, Furniture 17.2%, Grocery 13.1%

Verdict: FAIL TO REJECT $H_0 \rightarrow$ Age group and category preference are independent ($p=0.837$).

Business Translation:

The p-value of 0.837 is extremely high, meaning there is an 83.7% probability these observed category distributions within each age group occurred by random chance. Cramér's V of 0.051 confirms negligible association strength. While 26-35 and 46-55 age groups show slightly higher Electronics preference (~37.6–37.9% vs. 56-65 at 31.8%), these variations are well within statistical noise. Age is NOT a meaningful predictor of category choice.

All five age groups follow nearly identical category preferences: Electronics ~35% (dominant), Education ~15–20%, and Fashion/Furniture/Grocery each ~13–18%. A customer's age tells you almost nothing about whether they'll buy Electronics, Education, or Furniture.

Recommendation: Do NOT create age-targeted category campaigns. Scrap any plans for "Electronics for 26-35 year-olds" or "Education for 56-65 year-olds" messaging. Instead, use universal category strategy:

1. **Electronics promotion is uniform** across all age bands (already dominant at ~35–38%)
2. **Category bundling** (Electronics + Education) should target all ages equally
3. **Reserve budget for RFM-driven actions:** At-Risk win-back (₹35M exposure), Potential Loyalist reactivation (₹41M), and Champions VIP (retention)

Consolidated Business Recommendations

Finding	Statistical Evidence	Marketing Implication	Business Priority
Gender → Spending	✗ Not statistically significant ($t = 0.6820, p = 0.495389$)	Use broadly consistent messaging rather than creating separate spending strategies based solely on gender.	Low
Age Group → Category	✗ No significant association ($\chi^2 = 10.5361, p = 0.837174$; Cramér's $V = 0.0513$)	A universal category strategy is supported; avoid assuming category preference based solely on age group.	Low
RFM Segments	Descriptive / Segmentation Analysis — not a statistical significance test	Use segment-specific strategies: Champions → loyalty, At Risk → win-back, Potential Loyalists → retention/conversion.	High
Electronics Dominance	Descriptive Finding: ₹50.78M (36.4% of total revenue)	Protect the strongest category while exploring cross-selling and category diversification.	High
Geographic AOV Variation	Descriptive Finding: observed AOV varies across City × Category combinations; no significance test performed.	Investigate higher-AOV city/category combinations before considering location-specific pricing or promotional strategies.	Medium

Both statistical tests confirm: Traditional demographic segmentation (gender, age) has **zero discriminative power** in ApexPlanet's sales data. Gender explains 0% of AOV variance; age explains 0.26% of category choice variance.

Marketing Budget Reallocation (Immediate Actions):

1. ✗ Cancel gender-based A/B tests (they will show null results)
2. ✗ Eliminate age-specific category promotions (waste of creative/media spend)
3. ✓ **Focus 100% of segmentation budget on RFM lifecycle** (proven ₹35M–₹41M revenue exposure by segment from Task 3)
4. ✓ **Invest in geographic AOV optimization** (₹98K–₹182K AOV variance across cities is real; statistically sound)
5. ✓ **Category-level bundling** (Electronics + adjacent categories) applied uniformly to all demographics

Expected ROI Improvement: Eliminating gender/age-based campaigns and reallocating to RFM-driven retention/reactivation will yield 3–5x higher conversion rates vs. demographic spray-and-pray strategies (industry benchmark for e-commerce: RFM-based campaigns 8–12% response vs. demographic 2–3%).

Python Script

The full python script (sections 1–8 combined) is below and provided separately as ApexPlanet_Task4_Python_EDA.py.

```
# ApexPlanet Sales Analysis – Task 4: Hypothesis Testing
# Import libraries
import pandas as pd
import numpy as np
from scipy import stats
from scipy.stats import shapiro, levene, ttest_ind, chi2_contingency
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')

# Set display options
pd.set_option('display.max_columns', None)
pd.set_option('display.float_format', lambda x: f'{x:,.2f}' if abs(x) >= 1 else f'{x:.4f}')
sns.set_style('whitegrid')
plt.rcParams['figure.figsize'] = (12, 6)

# Load dataset
df = pd.read_csv('sales_dataset_python_analysis.csv')

# Data validation
print("=" * 70)
print("DATASET OVERVIEW")
print("=" * 70)
print(f"Shape: {df.shape[0]} rows × {df.shape[1]} columns")
print(f"Date Range: {df['Order_Date'].min()} to {df['Order_Date'].max()}")
print(f"\nTotal Revenue: ₹{df['Total_Sales'].sum():,.2f}")
print(f"Mean Order Value: ₹{df['Total_Sales'].mean():,.2f}")
print(f"Std Dev (Total_Sales): ₹{df['Total_Sales'].std():,.2f}")
print(f"\nGender Distribution: {df['Gender'].value_counts().to_dict()}")
print(f"Age_Group Distribution: \n{df['Age_Group'].value_counts().sort_index()}")
print(f"Category Distribution: \n{df['Category'].value_counts()}")
print(f"\nMissing Values: {df.isnull().sum().sum()}")
print("=" * 70)

print("\n" + "=" * 70)
print("HYPOTHESIS 1: GENDER DIFFERENCE IN MEAN TOTAL_SALES")
print("=" * 70)

# Separate data by gender
male_sales = df[df['Gender'] == 'Male']['Total_Sales']
female_sales = df[df['Gender'] == 'Female']['Total_Sales']

print(f"\nSample Sizes:")
print(f"  Male:    n = {len(male_sales)}")
print(f"  Female: n = {len(female_sales)}")

# Descriptive statistics
print(f"\n--- DESCRIPTIVE STATISTICS ---")
print(f"\nMale Customers:")
print(f"  Mean:      ₹{male_sales.mean():,.2f}")
print(f"  Median:    ₹{male_sales.median():,.2f}")
print(f"  Std Dev:   ₹{male_sales.std():,.2f}")
print(f"  Min / Max: ₹{male_sales.min():,.2f} / ₹{male_sales.max():,.2f}")

print(f"\nFemale Customers:")
print(f"  Mean:      ₹{female_sales.mean():,.2f}")
print(f"  Median:    ₹{female_sales.median():,.2f}")
print(f"  Std Dev:   ₹{female_sales.std():,.2f}")
print(f"  Min / Max: ₹{female_sales.min():,.2f} / ₹{female_sales.max():,.2f}")

print(f"\nDifference in Means:")
print(f"  Male - Female: ₹{male_sales.mean() - female_sales.mean():,.2f}")
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# Test assumptions
print(f"\n--- ASSUMPTION TESTS ---")

# Normality test (Shapiro-Wilk)
_, p_shapiro_male = shapiro(male_sales)
_, p_shapiro_female = shapiro(female_sales)
print(f"\nShapiro-Wilk Normality Test:")
print(f"  Male (p-value): {p_shapiro_male:.6f} {'✓ Normal' if p_shapiro_male > 0.05 else 'X Non-normal'}")
print(f"  Female (p-value): {p_shapiro_female:.6f} {'✓ Normal' if p_shapiro_female > 0.05 else 'X Non-normal'}")

# Homogeneity of variance (Levene's test)
_, p_levene = levene(male_sales, female_sales)
print(f"\nLevene's Test (Homogeneity of Variance):")
print(f"  p-value: {p_levene:.6f} {'✓ Equal variances' if p_levene > 0.05 else 'X Unequal variances'}")

# Independent t-test (use Welch's if variances unequal)
equal_var = p_levene > 0.05
t_stat, p_value_ttest = ttest_ind(male_sales, female_sales, equal_var=equal_var)

print(f"\n--- INDEPENDENT T-TEST ---")
print(f"Test Type: {'Student t-test' if equal_var else 'Welch t-test'} (equal_var={equal_var})")
print(f"  t-statistic: {t_stat:.4f}")
print(f"  p-value: {p_value_ttest:.6f}")
print(f"  α (sig level): 0.05")
print(f"  Decision: {'REJECT H0 ✓ Significant difference' if p_value_ttest < 0.05 else 'FAIL TO REJECT H0 → No significant difference'}")

# 95% Confidence Interval for difference in means
se_diff = np.sqrt((male_sales.var()/len(male_sales)) + (female_sales.var()/len(female_sales)))
mean_diff = male_sales.mean() - female_sales.mean()
ci_lower = mean_diff - 1.96 * se_diff
ci_upper = mean_diff + 1.96 * se_diff

print(f"\n95% Confidence Interval for Difference in Means:")
print(f"  Point Estimate: ₹{mean_diff:,.2f}")
print(f"  95% CI: [₹{ci_lower:,.2f}, ₹{ci_upper:,.2f}]")
print(f"  Interpretation: {'Zero is NOT in CI → significant' if (ci_lower > 0 or ci_upper < 0) else 'Zero is in CI → not significant'}")

# Visualization
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

# Box plot
axes[0].boxplot([male_sales, female_sales], labels=['Male', 'Female'], patch_artist=True)
axes[0].set_ylabel('Total Sales (₹)', fontsize=11)
axes[0].set_title('H1: Total Sales by Gender (Box Plot)', fontsize=12, fontweight='bold')
axes[0].grid(axis='y', alpha=0.3)
for patch, color in zip(axes[0].artists, ['#4FA8D8', '#E75A84']):
    patch.set_facecolor(color)
    patch.set_alpha(0.7)

# KDE plot
male_sales.plot(kind='kde', ax=axes[1], label='Male', linewidth=2.5, color='#4FA8D8')
female_sales.plot(kind='kde', ax=axes[1], label='Female', linewidth=2.5, color='#E75A84')
axes[1].set_xlabel('Total Sales (₹)', fontsize=11)
axes[1].set_ylabel('Density', fontsize=11)
axes[1].set_title('H1: Total Sales Distribution by Gender (KDE)', fontsize=12, fontweight='bold')
axes[1].legend(fontsize=10)
axes[1].grid(alpha=0.3)

plt.tight_layout()
plt.savefig('hypothesis1_gender_analysis.png', dpi=300, bbox_inches='tight')
print("\n✓ Visualization saved: hypothesis1_gender_analysis.png")
plt.show()

print("=" * 70)

print("\n" + "=" * 70)
print("HYPOTHESIS 2: AGE_GROUP & CATEGORY INDEPENDENCE")

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print("=" * 70)

# Create contingency table
contingency_table = pd.crosstab(df['Age_Group'], df['Category'])

print(f"\nContingency Table (Observed Frequencies):")
print(contingency_table)

# Calculate row and column totals
print(f"\n--- MARGIN TOTALS ---")
print(f"Row Totals (by Age_Group):")
print(contingency_table.sum(axis=1))
print(f"Column Totals (by Category):")
print(contingency_table.sum(axis=0))
print(f"Grand Total: {contingency_table.sum().sum()}")

# Chi-square test
chi2_stat, p_value_chi2, dof, expected_freq = chi2_contingency(contingency_table)

print(f"\n--- CHI-SQUARE TEST OF INDEPENDENCE ---")
print(f"   $\chi^2$  statistic:      {chi2_stat:.4f}")
print(f"  p-value:              {p_value_chi2:.6f}")
print(f"  Degrees of Freedom: {dof}")
print(f"   $\alpha$  (sig level):    0.05")
print(f"  Decision: {'REJECT  $H_0$  ✓ Significant association' if p_value_chi2 < 0.05 else 'FAIL TO REJECT  $H_0$  → No significant association'}")

# Expected frequencies
print(f"\nExpected Frequencies (if independent):")
expected_df = pd.DataFrame(expected_freq,
                           index=contingency_table.index,
                           columns=contingency_table.columns)
print(expected_df.round(2))

# Calculate standardized residuals
standardized_residuals = (contingency_table - expected_df) / np.sqrt(expected_df)
print(f"\nStandardized Residuals (|z| > 2 indicates strong deviation):")
print(standardized_residuals.round(2))

# Effect size (Cramér's V)
n = contingency_table.sum().sum()
min_dim = min(contingency_table.shape[0] - 1, contingency_table.shape[1] - 1)
cramers_v = np.sqrt(chi2_stat / (n * min_dim))
print(f"\nEffect Size (Cramér's V): {cramers_v:.4f}")
print(f"  Interpretation: {'Negligible' if cramers_v < 0.1 else 'Small' if cramers_v < 0.3 else 'Medium' if cramers_v < 0.5 else 'Large'} effect")

# Normalized contingency table (row percentages)
normalized_table = contingency_table.div(contingency_table.sum(axis=1), axis=0) * 100

print(f"\n--- NORMALIZED TABLE (% within each Age_Group) ---")
print(normalized_table.round(1))

# Visualization: Stacked bar chart
fig, ax = plt.subplots(figsize=(12, 6))

# Color palette
colors = ['#4FA8D8', '#E75A84', '#F5C547', '#8B6BA8', '#F2545B']
normalized_table.plot(kind='bar', stacked=True, ax=ax, color=colors, alpha=0.85, width=0.7)

ax.set_xlabel('Age Group', fontsize=11, fontweight='bold')
ax.set_ylabel('Proportion (%)', fontsize=11, fontweight='bold')
ax.set_title('H2: Category Preference by Age Group (Normalized Stacked Bar)', fontsize=12, fontweight='bold')
ax.legend(title='Category', bbox_to_anchor=(1.05, 1), loc='upper left', fontsize=9)
ax.set_ylim(0, 100)
ax.grid(axis='y', alpha=0.3)
plt.xticks(rotation=0)
plt.tight_layout()
plt.savefig('hypothesis2_age_category_analysis.png', dpi=300, bbox_inches='tight')
print("\n✓ Visualization saved: hypothesis2_age_category_analysis.png")
plt.show()

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print("=" * 70)

print("\n" + "=" * 70)
print("STATISTICAL VERDICTS & BUSINESS TRANSLATION")
print("=" * 70)

# Hypothesis 1 verdict
print(f"\n--- HYPOTHESIS 1 VERDICT ---")
h1_significant = p_value_ttest < 0.05
print(f"Statistical Finding:")
print(f"  Test Statistic (t): {t_stat:,.4f}")
print(f"  P-value: {p_value_ttest:.6f}")
print(f"  Significance Level:  $\alpha$  = 0.05")
print(f"  Conclusion: {'REJECT H0' if h1_significant else 'FAIL TO REJECT H0'})")

if h1_significant:
    direction = "Male" if male_sales.mean() > female_sales.mean() else "Female"
    print(f"\n ✓ STATISTICALLY SIGNIFICANT DIFFERENCE DETECTED")
    print(f" {direction} customers spend significantly more per order (p < 0.05)")
    print(f"  Mean difference: ₹{mean_diff:,.2f} [95% CI: ₹{ci_lower:,.2f} to ₹{ci_upper:,.2f}]")
else:
    print(f"\n X NO STATISTICALLY SIGNIFICANT DIFFERENCE")
    print(f"  Apparent spending difference (₹{mean_diff:,.2f}) is likely due to random variation")
    print(f"  95% CI includes zero: [₹{ci_lower:,.2f}, ₹{ci_upper:,.2f}]")

print(f"\n Business Implication:")
if h1_significant:
    print(f"  → Gender-based campaign targeting MAY be justified")
    print(f"  → Recommend A/B testing gender-specific messaging on 10% sample")
    print(f"  → Budget allocation by gender conversion funnel metrics (not AOV alone)")
else:
    print(f"  → Gender-based AOV targeting NOT justified")
    print(f"  → Recommend RFM/Category segmentation instead of gender splits")
    print(f"  → Unified messaging for both genders; optimize by lifecycle stage")

# Hypothesis 2 verdict
print(f"\n--- HYPOTHESIS 2 VERDICT ---")
h2_significant = p_value_chi2 < 0.05
print(f"Statistical Finding:")
print(f"  Test Statistic ( $\chi^2$ ): {chi2_stat:,.4f}")
print(f"  P-value: {p_value_chi2:.6f}")
print(f"  DF: {dof}")
print(f"  Significance Level:  $\alpha$  = 0.05")
print(f"  Cramér's V (effect): {cramers_v:.4f} ({'Negligible' if cramers_v < 0.1 else 'Small' if cramers_v < 0.3 else 'Medium' if cramers_v < 0.5 else 'Large'})")
print(f"  Conclusion: {'REJECT H0' if h2_significant else 'FAIL TO REJECT H0'})")

if h2_significant:
    print(f"\n ✓ STATISTICALLY SIGNIFICANT ASSOCIATION DETECTED")
    print(f"  Age group and category preference are NOT independent (p < 0.05)")
    # Identify strongest associations
    max_residuals = standardized_residuals.abs().unstack().nlargest(5)
    print(f"\n Strongest Associations (Top 5 by |residual|):")
    for idx, (age_cat, residual_val) in enumerate(max_residuals.items(), 1):
        age, category = age_cat
        actual_pct = (contingency_table.loc[age, category] / contingency_table.loc[age].sum()) * 100
        print(f"  {idx}. Age {age} + {category}: {actual_pct:.1f}% (residual: {standardized_residuals.loc[age, category]:.2f})")
    else:
        print(f"\n X NO STATISTICALLY SIGNIFICANT ASSOCIATION")
        print(f"  Age group distribution appears uniform across categories")
        print(f"  Observed variations (if any) are consistent with random chance")

print(f"\n Business Implication:")
if h2_significant:
    print(f"  → Age-targeted category promotions ARE data-justified")
    print(f"  → Create age-specific product recommendations (e.g., 'Trending in 36-45 group')")
    print(f"  → Allocate email/social inventory per age-category combo with highest demand")
    print(f"  → Electronics shows different appeal by age – investigate messaging refinement")

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else:
    print(f"    → Age-targeted category campaigns NOT justified by data")
    print(f"    → Use uniform category strategy across all age bands")
    print(f"    → Segment instead by Recency/Monetary (RFM) for higher ROI")

# Summary table
print(f"\n--- SUMMARY OF STATISTICAL TESTS ---")
summary_data = {
    'Hypothesis': ['H1: Gender → Spending', 'H2: Age_Group ↔ Category'],
    'Test': ['Independent t-test', 'Chi-Square ( $\chi^2$ )'],
    'Test Statistic': [f'{t_stat:,.4f}', f'{chi2_stat:,.4f}'],
    'P-value': [f'{p_value_ttest:.6f}', f'{p_value_chi2:.6f}'],
    'α (threshold)': ['0.05', '0.05'],
    'Verdict': ['REJECT' if h1_significant else 'FAIL TO REJECT', 'REJECT' if h2_significant else 'FAIL TO REJECT'],
    'Significant?': ['YES ✓' if h1_significant else 'NO X', 'YES ✓' if h2_significant else 'NO X']
}

summary_df = pd.DataFrame(summary_data)
print(summary_df.to_string(index=False))

print("\n" + "=" * 70)
print("END OF ANALYSIS")
print("=" * 70)

```